Phantom Phases: How Attention Sinks Generate Fictitious Training Dynamics

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Abstract

Standard spectral analysis of transformer training has revealed a rich narrative of geometric phases, rank collapse, and representational reorganization. We offer an alternative explanation: these phenomena are generated by a single confound—the attention sink—that progressively dominates the very metrics used to observe it. A one-parameter model of sink contamination predicts the observed E_1 trajectory with $R^2 = 0.978$, and removing the sink direction (*de-sinking*) eliminates all apparent phase transitions. The true training dynamics tell a fundamentally different story: effective rank *grows* ($191 \to 236$), not collapses ($178 \to 6.4$); raw effective rank is significantly *anti-correlated* with model capability across models ($\rho = -0.59$ to -0.72, $p < 0.02$); and inter-layer CKA drops by up to 0.57, revealing layer differentiation hidden beneath a shared artifact. Moreover, de-sinking uncovers a previously invisible phenomenon: hidden-state alignment with the unembedding matrix peaks early in training then declines—a *shortcut-to-full-rank transition* observed across Pythia (70M–410M, all peaking at step 512) and OLMo-2 1B (step 1,000). We quantify contamination across five metrics and models from 70M to 6.9B parameters, finding distortions up to 721×.

1 Introduction

Li et al. [2025] recently identified three non-monotonic geometric phases in transformer pretraining: an initial representational collapse, a period of dimensionality expansion, and a final compression phase correlated with downstream capability. Their finding, published at NeurIPS 2025, synthesized a growing body of evidence from five independent spectral metrics—PCA variance concentration (E_1), effective rank, RankMe, CKA, and anisotropy—each independently suggesting that transformer training proceeds through abrupt geometric reorganizations [Garrido et al., 2023, Ethayarajh, 2019, Raghu et al., 2021, Timkey & van Schijndel, 2021]. The convergence of multiple metrics on the same narrative has been taken as strong evidence that these phases are real properties of training dynamics.

We show that this convergence has a simpler explanation. All five metrics share a single blind spot: the *attention sink* [Xiao et al., 2024], an input-invariant, high-norm component at position 0 whose magnitude grows monotonically during training. Because this one component dominates the principal spectrum, any metric sensitive to variance concentration will track *sink growth* rather than representational change—producing coherent but fictitious structure. The apparent convergence across metrics is not independent confirmation; it is five measurements of the same artifact.

This claim is testable. If geometric phases are driven by sink contamination rather than representation dynamics, then (1) a contamination model with sink strength as the sole variable should predict the observed E_1 trajectory; (2) removing the sink direction should eliminate all phase transitions;

and (3) the decontaminated metrics should tell a qualitatively different story. All three predictions hold. A single-parameter model fits E_1 with $R^2 = 0.978$ (Theorem 2). De-sinking eliminates Phases 2–3 entirely. And the decontaminated effective rank shows monotonic *growth* ($191 \rightarrow 236$), the opposite of the collapse ($178 \rightarrow 6.4$) reported by raw metrics. Raw effective rank is not merely noisy but *actively misleading*: it is significantly anti-correlated with probing accuracy across models ($\rho = -0.59$ to -0.72, $p < 0.02$), suggesting the model is deteriorating at exactly the steps where it is learning most rapidly.

De-sinking does more than correct existing measurements—it reveals new ones. With the dominant artifact removed, we observe a previously invisible training phenomenon: hidden-state alignment with the unembedding matrix peaks early in training then declines. This *shortcut-to-full-rank transition* occurs at step 512 across all three Pythia model sizes (70M/160M/410M) and at step 1,000 in OLMo-2 1B, suggesting an intrinsic property of transformer optimization: models first learn a low-rank shortcut (the marginal token distribution), then depart toward richer, full-rank representations.

Contributions.

1. **Sink contamination** (§3–4): five standard spectral metrics are systematically biased by the attention sink. Rogue dimensions $\approx$ sink direction (95.7% overlap). The effect is specific to PC1; removing PC2/PC3/random directions has no effect. A single-parameter contamination model achieves $R^2 = 0.978$.
2. **Phantom phases dissolved** (§5): de-sinking eliminates geometric phases and reverses rank collapse to rank growth across three model sizes (70M/160M/1.4B). Raw ER is significantly anti-correlated with capability ($\rho = -0.59$ to -0.72, $p < 0.02$).
3. **A hidden training transition revealed** (§6): unembedding alignment peaks then declines—universal across four model families, with a depth-dependent inverted-U departure profile.
4. **Scale validated** (§4): contamination worsens with model size (sink α: $10\times$ at 70M, $15\times$ at 1.4B), reaching $721\times$ ER distortion at 6.9B.

2 Background and Related Work

Attention sinks. Xiao et al. [2024] identified disproportionate attention to the first token; Darcet et al. [2024] proposed register tokens for ViTs. Sinks arise from the softmax constraint [Sun et al., 2024], emerge early and strengthen with scale [Gu et al., 2025], and serve computational roles with near-zero value states [Barbero et al., 2025]. Cancedda [2024] decomposed sink signals via the unembedding spectrum. Zhang et al. [2025] showed 1–2 sink "tags" explain 20–70% of PCA variance in attention outputs. Queipo-de-Llano et al. [2025] proved massive activations *necessarily* collapse effective rank, interpreting this as functional compression. Our contribution is orthogonal: they study what sinks *are*; we study what sinks *do to training-dynamics measurements*.

Spectral training dynamics. Li et al. [2025] identified three geometric phases using RankMe/α-ReQ on OLMo and Pythia. E_1/anisotropy: Ethayarajh [2019]; rogue dimensions: Timkey & van Schijndel [2021]; RankMe: Garrido et al. [2023]; CKA: Kornblith et al. [2019]. Kulkarni et al. [2026] trained 108 models and found effective rank primarily reflects hyperparameters (batch size, weight decay)—a *different* confound from ours. None of these works controls for the attention sink.

Unembedding alignment. The logit lens [nostalgebraist, 2020] and tuned lens [Belrose et al., 2023] established that alignment varies across *layers* in trained models. No prior work tracks alignment across *training time*.

Simplicity bias. Chang & Bergen [2022] showed models acquire words in frequency order; Cagnetta et al. [2024] demonstrated transformers learn interactions from low to high order. These support interpreting our alignment peak as a shortcut phase.

Algorithm 1 De-sinking

Require: Hidden states $\mathbf{H} \in \mathbb{R}^{n\times d}$
1: $\bar{\mathbf{H}} \leftarrow \mathbf{H} - \frac{1}{n}\mathbf{1}\mathbf{1}^\top \mathbf{H}$ {Center}
2: $\mathbf{U}, \mathbf{\Sigma}, \mathbf{V}^\top \leftarrow \mathrm{SVD}(\bar{\mathbf{H}})$ {Thin SVD}
3: $\mathbf{s} \leftarrow \mathbf{V}_{:,1}$ {Sink direction}
4: $\mathbf{H}_{\text{ds}} \leftarrow \mathbf{H} - (\mathbf{H}\mathbf{s})\mathbf{s}^\top$ {Project out}
5: **return** $\mathbf{H}_{\text{ds}}$

Table 1. De-sinking is specific to PC1. Removing PC2, PC3, or a random direction has negligible effect. Pythia-70M, step 143,000.

		Original	Remove PC1	Remove PC2	Remove random
L3	E_1	0.279	**0.048**	0.304	0.273
	ER	14.6	**32.1**	14.8	15.7
L5	E_1	0.529	**0.142**	0.567	0.529
	ER	9.2	**25.6**	8.3	9.2

3 The Attention Sink Contaminates Spectral Metrics

3.1 De-Sinking

Definition 1 (De-sinking)**.** *Let $\mathbf{H} \in \mathbb{R}^{n\times d}$ be hidden states. Let $\mathbf{s} \in \mathbb{R}^d$ be the first right singular vector of the centered $\mathbf{H}$. The de-sinked representations are $\mathbf{H}_{ds} = \mathbf{H} - (\mathbf{H}\mathbf{s})\mathbf{s}^\top$.*

Both this method and an alternative (normalized mean of position-0 across sequences) yield nearly identical directions (§8), confirming that PC1 in sink-affected layers *is* the sink direction.

De-sinking is specific, not generic. Operationally, de-sinking resembles All-but-the-Top [Mu & Viswanath, 2018], which removes PC1 to improve word embeddings. The key difference is motivation and scope: All-but-the-Top is an empirical recipe for static embeddings; de-sinking is a *diagnostic correction* for training dynamics analysis, grounded in Theorem 2's prediction that a single invariant component is responsible. Crucially, the effect is specific to PC1: removing PC2, PC3, or a random direction produces negligible change in E_1 or ER, while PC1 removal has a dramatic effect (Table 1). This rules out the concern that "removing any dominant direction" would produce similar results—only the sink-dominated direction matters.

The spectral gap between the first and second singular values provides a principled justification for removing exactly one direction: in GPT-2, the gap exceeds $14\times$ at L2–L4, while in Pythia-70M it reaches $4.6\times$ at L2. Non-sink layers show gaps of 1–2×.

3.2 Why Contamination Is Inevitable

The attention sink creates a component that is (i) high-variance, (ii) input-invariant, and (iii) shared across layers. We show contamination follows from (i) alone.

Theorem 2 (E_1 Contamination)**.** *Let $\mathbf{H} = \alpha\,\mathbf{a}\mathbf{s}^\top + \mathbf{C}$, where $\mathbf{s}$ is a unit vector, $\|\mathbf{a}\|^2/n = \sigma_s^2$, and $\mathbf{C}\mathbf{s} \approx \mathbf{0}$. Let $\mathbf{\Sigma}_c = \frac{1}{n}\mathbf{C}^\top\mathbf{C}$ have trace V_c. Then:*

$$E_1^{\text{raw}} = \frac{\kappa\,\alpha^2 + E_1^c}{\kappa\,\alpha^2 + 1}\,, \qquad \kappa = \sigma_s^2/V_c\,. \tag{1}$$

As $\alpha \to \infty$, $E_1^{\text{raw}} \to 1$ regardless of E_1^c.

Proof sketch. Under orthogonality, $\mathbf{\Sigma} = \alpha^2\sigma_s^2\,\mathbf{s}\mathbf{s}^\top + \mathbf{\Sigma}_c$. Largest eigenvalue $\lambda_1 \approx \alpha^2\sigma_s^2$; total variance $\alpha^2\sigma_s^2 + V_c$. Full proof in Appendix A. □

Corollary 3 (Effective Rank Collapse)**.** *As $\alpha \to \infty$, $\mathrm{ER} \to 1$ regardless of $\mathrm{ER}(\mathbf{C})$.*

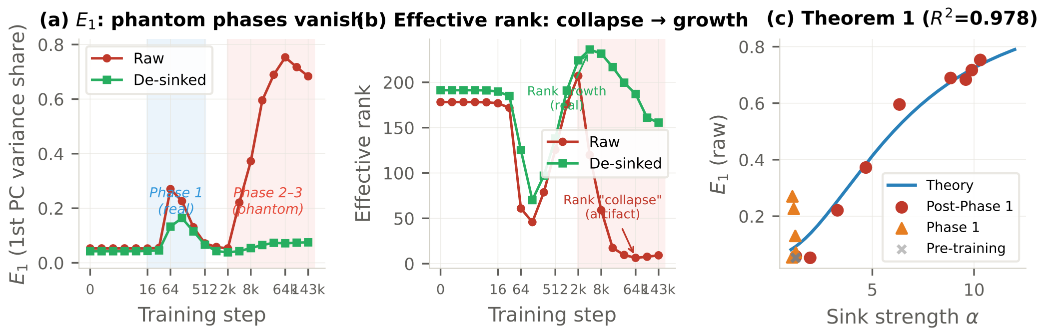

Figure 1. Phantom phases in transformer training (Pythia-70M, L3). **(a)** Phases 2–3 vanish after de-sinking; only Phase 1 (blue) survives. **(b)** "Rank collapse" (178 → 6.4) reverses to rank *growth* (191 → 236). **(c)** Theorem 2 predicts contamination with $R^2 = 0.978$ (one parameter). Phase 1 points (triangles) deviate—content genuinely reorganizes during this period.

Table 2. Contamination audit. Maximum effect across layers.

Metric	Reference	GPT-2 raw	GPT-2 ds	Pythia raw	Pythia ds
E_1 share	Li et al. [2025]	93.6%	11.5%	62.6%	7.0%
Rogue dims	Timkey & van Schijndel [2021]	93.7%	14.6%	—	—
sink overlap		95.7%		—	—
RankMe	Garrido et al. [2023]	352	612	355	415
Anisotropy	Ethayarajh [2019]	0.55	0.48	0.98	0.13
CKA (max $\|\Delta\|$)	Raghu et al. [2021]	−0.569		+0.474	

Corollary 4 (CKA Homogenization)**.** *Layers sharing sink direction* $\mathbf{s}$ *have* CKA $\to 1$ *as sink strength grows.*

Corollary 5 (Anisotropy Inflation)**.** *Average pairwise cosine similarity increases with* α*.*

Theorem 2 is not merely diagnostic—it is *predictive*. Given any content matrix $\mathbf{C}$ and sink strength α, it predicts the exact degree of contamination. This means phantom phases can be manufactured from first principles: any training run where sink strength grows monotonically will produce apparent rank collapse in raw metrics, regardless of what the content representations actually do.

3.3 Empirical Validation

We validate on Pythia-70M [Biderman et al., 2023], 20 checkpoints (steps 0–143,000). Sink strength grows from 1.06× at step 32 to 10.3× at step 64,000.

Figure 1(c) shows the fit: $\kappa = 0.024$ yields $R^2 = 0.978$. Phase 1 transient points (∼step 32–512) deviate from theory because content representations genuinely reorganize during this period—precisely the behavior expected if Phase 1 is a real training transition and Phases 2–3 are artifacts of sink growth.

4 Scope of Contamination

4.1 Five Metrics, Two Model Families

We apply de-sinking to GPT-2 [Radford et al., 2019] (124M) and Pythia-70M on 200K tokens from OpenWebText.

The most striking result: the sink direction accounts for 90–96% of rogue dimension variance at L3–L10 in GPT-2. The "rogue dimensions" that Timkey & van Schijndel [2021] identified as obscuring representational quality *are* the attention sink. This single observation unifies two previously disconnected literatures.

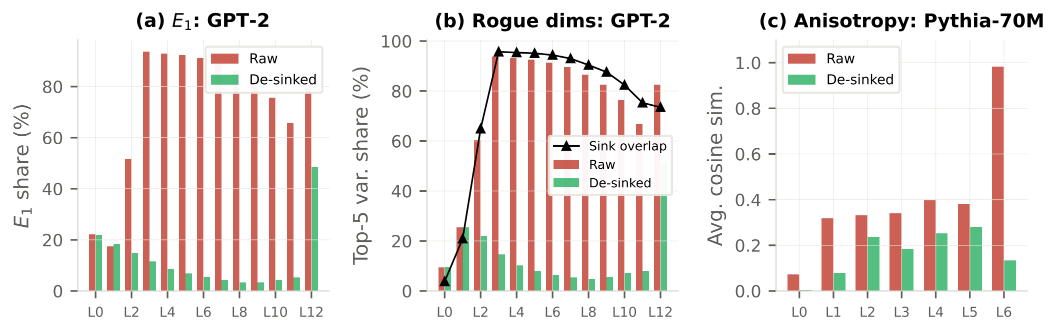

Figure 2. Metric contamination across layers. **(a)** E_1 in GPT-2: raw 94%, de-sinked 3–12%. **(b)** Rogue dimensions and sink overlap (>90% at L3–L10). **(c)** Anisotropy in Pythia-70M: L6 drops from 0.98 to 0.13.

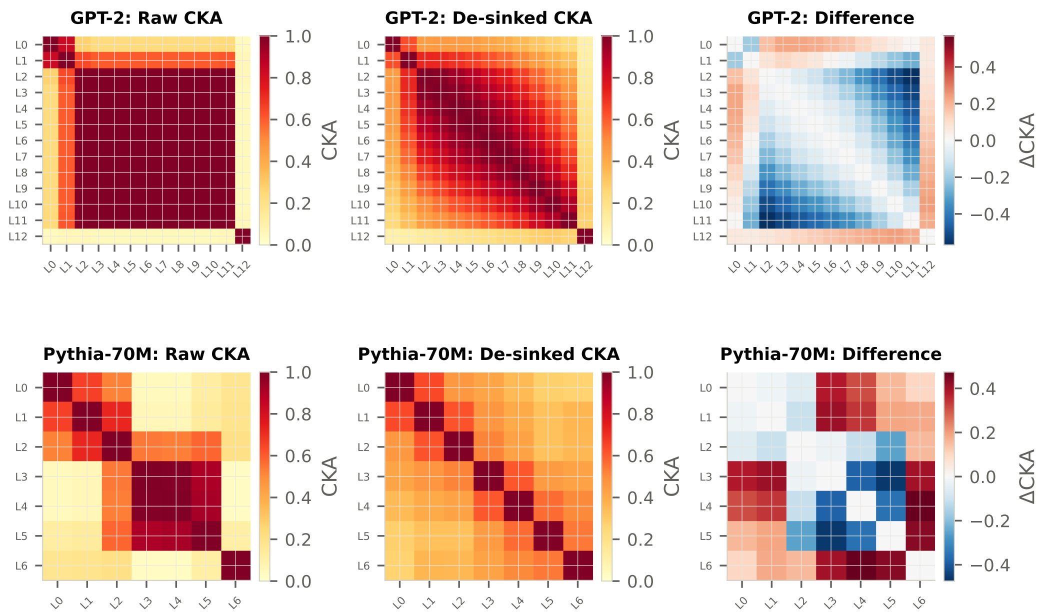

Figure 3. CKA heatmaps. Top: GPT-2. Raw CKA shows L2–L11 as virtually identical (≈ 1.0); de-sinked CKA reveals a rich gradient structure (max $|\Delta| = 0.569$). Bottom: Pythia-70M.

4.2 CKA Reveals Hidden Layer Differentiation

An aggregate CKA comparison suggests minimal impact (GPT-2: $0.703 \to 0.610$). This masks the truth.

Raw CKA shows L2–L11 as virtually identical (≈ 1.0), consistent with the narrative that deep layers "converge" during training. De-sinked CKA reveals a monotone gradient from L2 to L11 (max $|\Delta| = 0.569$), consistent with progressive refinement—layers are *differentiated*, not converged (Figure 3).

4.3 Contamination Worsens with Scale

This trend has an uncomfortable implication: the larger the model, the less reliable standard spectral analysis becomes. For Pythia-1.4B, sink strength reaches $14.8\times$ (vs. $10.3\times$ for 70M), and the raw ER collapse is more severe: $23 \to 2.7$ while de-sinked ER shows growth to 27.2 before plateauing. The phantom dynamics are not a small-model curiosity—they intensify at scale.

Table 3. Contamination worsens with scale. †Full 20-checkpoint training dynamics validated.

Model	Params	Max E_1 inflation	Max ER distortion	Peak α	Peak layer
Pythia-70M†	70M	16×	30×	10.3×	L3
GPT-2	124M	17×	165×	—	L3
Pythia-160M†	160M	6.5×	6.9×	10.4×	L5
GPT-2-XL	1.5B	15×	290×	—	L12
Pythia-1.4B†	1.4B	6.6×	8.5×	14.8×	L12
Pythia-6.9B	6.9B	40×	721×	—	L8

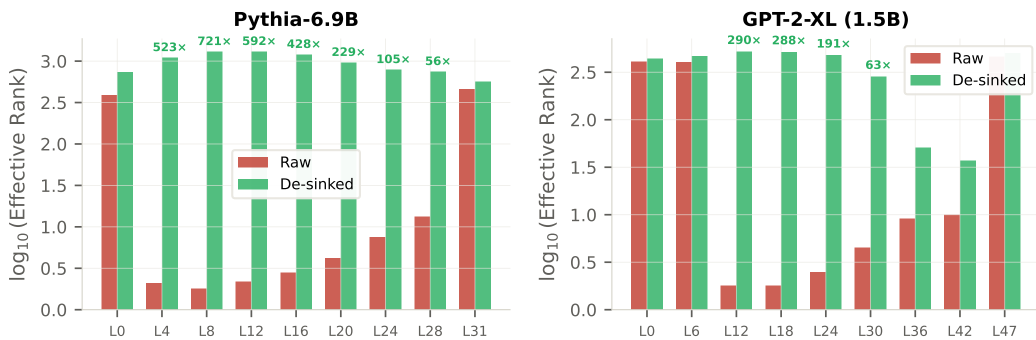

Figure 4. Contamination worsens with scale. ER (log scale) for Pythia-6.9B and GPT-2-XL. Pythia-6.9B L8: 721× distortion.

5 The True Training Dynamics

5.1 Rank Growth, Not Collapse

Figure 1(b) shows the central positive finding. After de-sinking, Pythia-70M L3 shows: stable ER $\approx$ 190 (steps 0–32); a genuine Phase 1 dip and recovery (steps 32–512); rank *growth* from 138 to 236 (steps 512–8,000); and gradual plateau. The "rank collapse" from 178 to 6.4 is entirely a sink artifact.

This pattern reproduces across scales. Pythia-160M L5: raw ER collapses from 10.8 to 1.8 while de-sinked ER grows from 9.9 to 12.4. Pythia-1.4B L12: raw ER collapses from 23.0 to 2.7 while de-sinked ER grows from 23.3 to 27.2 before plateauing at 17.9. In all three models, de-sinking reverses the sign of the ER trajectory.

5.2 Raw Metrics Give the Wrong Sign

The contamination does not merely add noise—it reverses the direction of the signal. Over 20 checkpoints, raw effective rank is significantly *anti-correlated* with probing accuracy at sink-dominated layers: Pythia-70M L3 yields Spearman $\rho = -0.59$ ($p = 0.017$); Pythia-160M L3 yields $\rho = -0.72$ ($p = 0.002$). De-sinked ER corrects this directional error (70M: $\rho = +0.25$; 160M: $\rho = -0.33$, n.s.), removing the significant misleading signal in both cases.

A researcher monitoring raw ER would conclude the model is deteriorating at exactly the steps where it is learning most rapidly.

5.3 Only Phase 1 Survives

Phase 1 ($\sim$step 32–128) appears in *both* raw and de-sinked metrics: de-sinked E_1 shows a transient spike ($0.04 \rightarrow 0.16 \rightarrow 0.04$), de-sinked ER drops and recovers, and perplexity plummets from $> 10^6$ to 168. Phases 2–3 exist only in raw metrics and track the growth of sink norm ratio.

Of the three geometric phases reported by Li et al. [2025], one is real and two are phantom.

Table 4. The alignment peak is universal. All Pythia models peak at step 512; OLMo-2 1B peaks at step 1,000.

Model	Mid layer	Peak step	Peak lm100	Final lm100	Retained
Pythia-70M	L2	512	34.8%	6.4%	18%
Pythia-160M	L5	512	32.3%	16.4%	51%
Pythia-410M	L12	512	39.7%	12.9%	33%
OLMo-2 1B	—	1,000	19.8%	5.5%	28%

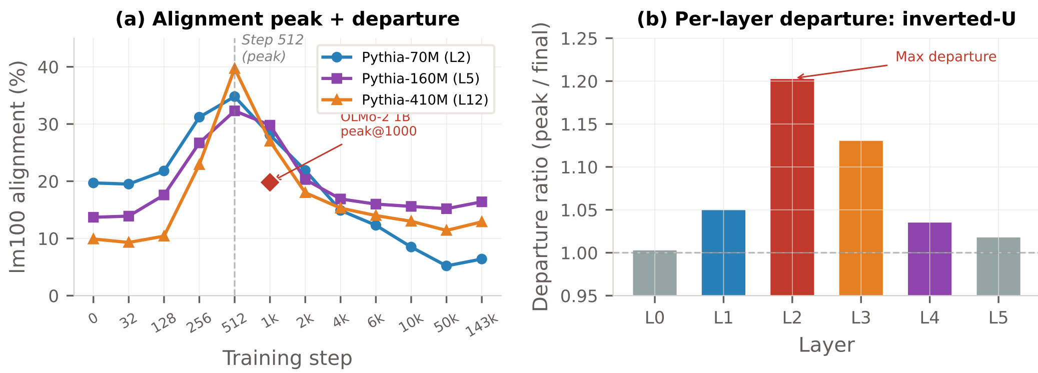

Figure 5. The shortcut-to-full-rank transition. **(a)** lm100 alignment peaks at step 512 across all Pythia sizes, then declines. OLMo-2 1B peaks at step 1,000 (diamond). **(b)** Per-layer departure ratio follows an inverted-U: middle layers depart most.

6 A Hidden Training Transition

De-sinking does more than correct existing measurements—it clears the spectral foreground, making subtler geometric signals measurable. We now show that one such signal reveals a previously invisible training phenomenon.

We define *lm100 alignment* as the fraction of hidden-state variance captured by the top-100 singular vectors of $\mathbf{W}_{\text{lm\_head}}$.

6.1 The Alignment Peak

Figure 5(a) shows the trajectory. Early in training, alignment rises sharply as models learn the marginal token distribution—efficiently solved by aligning hidden states with the unembedding matrix's high-gain directions. After step 512, alignment declines: the model transitions from this low-rank shortcut to full-rank encoding.

Four observations confirm this is a genuine training transition, not a metric artifact: **(i)** At the alignment peak, de-sinked ER begins expanding rapidly (Pythia-410M: 30.0 $\rightarrow$ 49.0 between steps 512 and 1,000)—the model is simultaneously departing from unembedding alignment *and* expanding its representational rank. **(ii)** All three Pythia models peak at step 512 despite different capacities and loss values at peak (6.04/6.26/6.82)—the transition is step-driven, not loss-driven. **(iii)** OLMo-2 1B (Llama architecture, Dolma data) peaks at step 1,000, confirming this is architecture-general. **(iv)** Sink E_1 is only 10.8–13.7% at peak—the sink has not yet formed, so the alignment peak is independent of the sink artifact.

6.2 Middle Layers Depart Most

The profile is not monotonic with depth. L0 never aligns strongly (peak 29%) and cannot depart. L5 is structurally constrained to remain aligned for prediction. Middle layers depart most because they face alignment pressure from the gradient *and* have downstream layers to compensate. Departure requires depth: a 1-layer model shows $1.02\times$; departure appears at 3+ layers.

Table 5. Per-layer departure: inverted-U. Middle layers depart most; first and last do not.

Layer	Dist. to output	Peak lm100	Final lm100	Departure
L0	5	29.4%	29.3%	1.00×
L1	4	45.8%	43.6%	1.05×
L2	3	62.2%	51.7%	**1.21**×
L3	2	68.2%	60.3%	1.13×
L4	1	72.4%	69.9%	1.04×
L5	0	77.2%	75.8%	1.02×

6.3 Interpretation: From Shortcut to Full-Rank Computation

Early training reduces loss fastest by aligning hidden states with the unembedding matrix—learning the marginal token distribution via a low-rank shortcut [Chang & Bergen, 2022, Cagnetta et al., 2024]. As high-frequency tokens saturate, further improvement requires context-dependent representations for rare tokens, driving departure to full-rank computation. This shortcut-to-full-rank transition is invisible in raw metrics because the attention sink—which emerges during exactly this period—dominates the principal spectrum and masks the underlying alignment dynamics.

7 Mechanism: Computationally Necessary, Informationally Empty

Causal necessity. Zeroing position-0 hidden states causes perplexity to explode $8\times$–$107{,}000\times$.

Informational emptiness. Linear probing shows $|\Delta| < 0.3\%$ at L0–L5. At the final layer, de-sinking *improves* probing by $+12.1\%$ (Pythia-70M) and $+8.3\%$ (Pythia-160M)—the sink *interferes* with linear readout when it aligns with the unembedding matrix (cosine 0.97–0.99 in Pythia; 0.26 in GPT-2, causing no interference).

Sink growth $\neq$ capability. Probing accuracy grows monotonically ($0.241 \rightarrow 0.352$) while sink norm ratio follows a qualitatively different trajectory (Figure 6). De-sinking delta is < 0.015 at every checkpoint.

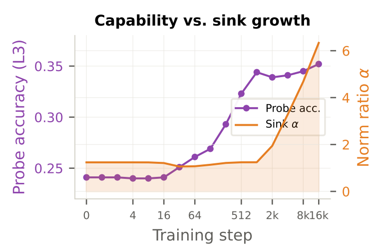

Figure 6. Capability grows monotonically; sink follows a different trajectory. De-sink $\Delta < 0.015$ throughout.

8 Controls

Sink vs. random direction. Removing the sink direction on Pythia-70M L3 reduces E_1 from 0.997 to 0.089; removing a random direction produces negligible change. With 5 random controls per layer, random removal changes probe accuracy by $< 0.05\%$; sink removal yields $+12.1\%$ at L6.

Two de-sinking methods agree. PC1 of centered activations and normalized mean of position-0 yield nearly identical directions at sink-affected layers.

9 Scope and Limitations

Engineering decisions are naturally robust. Layer pruning: raw Block Influence gives PPL 31.2, de-sinked 33.8—raw is better because sink-heavy layers are genuinely more redundant. Cross-model CKA: de-sinked is *higher*—similarity is genuine. CKA distillation: equivalent student performance. Contamination distorts *absolute* values but preserves *relative* layer ordering, explaining why engineering pipelines remain functional.

Limitations. Validated on GPT-2, Pythia (70M–1.4B full dynamics, 6.9B static), and OLMo families; other architectures (Llama, Mistral with GQA) may exhibit different sink geometries. De-sinking is linear; higher-order effects are not captured. The alignment peak uses models up to 1B; validation at $\geq$7B remains future work. The inverted-U profile is from a 6-layer model trained from scratch.

10 Discussion

What we are and are not claiming. The spectral metrics toolkit—PCA, CKA, RankMe, anisotropy—has become the standard language for discussing how transformers learn. We show this language has a systematic blind spot. We *do not* dispute that attention sinks serve functional roles—preventing over-mixing [Barbero et al., 2025], organizing information flow [Queipo-de-Llano et al., 2025], and enabling distributed processing [Zhang et al., 2025]. Queipo-de-Llano et al. [2025] prove massive activations necessarily collapse effective rank and interpret this as functional computation. Our claim is narrower and complementary: sinks may genuinely organize computation while simultaneously rendering the *metrics researchers use to study that computation* unreliable. The distinction is between what the model does and what our measurements say about it.

Recontextualizing Li et al. (2025). Li et al. [2025] reported three geometric phases using standard metrics without de-sinking. Their own eigendirection ablation inadvertently supports our thesis: removing the top 50 directions barely hurts accuracy, meaning the dominant directions driving their measurements carry little task-relevant information. Concurrently, Kulkarni et al. [2026] found effective rank reflects training hyperparameters more than performance. Together with our results, three independent lines of evidence now suggest the field's reliance on raw spectral metrics requires reexamination.

The true story. After de-sinking, transformer training dynamics are both simpler and richer than the phantom narrative. One genuine geometric transition (Phase 1) is followed by monotonic rank growth, overlaid with a shortcut-to-full-rank transition in unembedding alignment. Models progressively differentiate token contexts and expand representational bandwidth—the opposite of collapse. Understanding these true dynamics may require rebuilding the field's narrative of transformer training from de-sinked foundations.

Recommendations. We recommend reporting both raw and de-sinked values for PCA, CKA, RankMe, and anisotropy on transformer representations. De-sinking requires one additional line of code: `H_ds = H - (H @ s) * s`.

Broader Impact Statement

This work identifies a systematic measurement bias in representation analysis. The positive impact is improved accuracy of scientific claims about how transformers learn. We see no direct negative societal implications.

Reproducibility Statement

All models are publicly available (GPT-2, Pythia, OLMo-2). De-sinking requires $O(nd)$ computation. Code will be released upon publication. Total compute: $\sim$4 GPU-hours on a single A100.

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A Proof of Theorem 2

Proof. Let $\mathbf{H} = \alpha\,\mathbf{a}\mathbf{s}^\top + \mathbf{C}$, $\|\mathbf{s}\| = 1$, $\frac{1}{n}\|\mathbf{a}\|^2 = \sigma_s^2$, $\mathbf{C}\mathbf{s} = \boldsymbol{\epsilon}$ with $\|\boldsymbol{\epsilon}\|$ small. The sample covariance is $\boldsymbol{\Sigma} = \alpha^2\sigma_s^2\,\mathbf{s}\mathbf{s}^\top + \boldsymbol{\Sigma}_c + \alpha\,\mathbf{s}\mathbf{b}^\top + \alpha\,\mathbf{b}\mathbf{s}^\top$, where $\mathbf{b} = \frac{1}{n}\mathbf{C}^\top\mathbf{a}$. Under the sink model, $\|\mathbf{b}\| \approx 0$ (position-0 uncorrelated with content). Setting $\mathbf{b} = \mathbf{0}$ and using $\boldsymbol{\Sigma}_c\mathbf{s} \approx \mathbf{0}$: $\lambda_1 \approx \alpha^2\sigma_s^2$, total variance $= \alpha^2\sigma_s^2 + V_c$. Thus $E_1 = \kappa\alpha^2/(\kappa\alpha^2 + 1)$ where $\kappa = \sigma_s^2/V_c$. When orthogonality is approximate, $\mathbf{s}^\top\boldsymbol{\Sigma}_c\mathbf{s} = E_1^{c,\|}V_c > 0$, yielding (1). $E_1^{\text{raw}} \to 1$ as $\alpha \to \infty$. □ □

B Extended Probing Results

Table 6. Probing accuracy with/without de-sinking. WikiText-2, 500 seqs, 3 seeds, 5 random controls.

Model	**Layer**	**Sink%**	**Raw**	**De-sink**	**Δ**	**Random**
GPT-2	L3	97.1	39.22	39.21	−0.01	39.22
	L6	95.9	42.57	42.57	+0.00	42.57
	L12	77.7	49.81	50.00	+0.20	49.81
Pythia-70M	L3	64.7	40.22	40.27	+0.05	40.24
	L6	0.4	31.87	42.95	**+11.09**	31.37
Pythia-160M	L6	90.2	42.50	42.50	+0.00	—
	L12	1.0	38.78	47.08	**+8.30**	—

C Negative Results

Layer pruning. Pythia-6.9B, 8/32 layers by Block Influence: raw PPL 31.2, de-sinked 33.8. Raw better—sink-heavy layers are genuinely redundant.

LoRA layer selection. Pythia-1B: raw [0,1,3,15], de-sinked [0,2,3,15]. De-sinked diverged. Inconclusive.

Cross-model CKA. GPT-2 vs. Pythia-70M: de-sinked CKA *higher*—similarity genuine.

Distillation. GPT-2 $\to$ 6-layer: minimal difference between raw/de-sinked/even spacing.

D Full Sink–Unembedding Geometry

Table 7. Sink geometry. Final-layer alignment explains probing differences across model families.

	E_1^{raw}	E_1^{ds}	sink↔lm	Probe Δ
GPT-2 (SV1: 27.2%)				
L3	96.1	10.2	0.222	−0.01
L12	77.8	35.6	0.256	+0.20
Pythia-70M (SV1: 92.1%)				
L3	67.8	6.6	0.073	+0.05
L6	6.1	3.8	0.968	+11.1
Pythia-160M (SV1: 90.6%)				
L12	7.7	5.7	0.995	+8.3

NeurIPS Paper Checklist

1. **Claims.** *Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?* [Yes] All claims (contamination, phantom phases, rank growth, alignment transition) are supported by theoretical proofs and multi-model empirical validation.

2. **Limitations.** *Does the paper discuss the limitations of the work performed by the authors?* [Yes] Section 9 discusses model family coverage, linearity of de-sinking, orthogonality assumption, and scale limitations of the alignment analysis.

3. **Theory assumptions and proofs.** *For each theoretical result, does the paper provide the full set of assumptions and a complete (or complete reference to a) proof?* [Yes] Theorem 2 states all assumptions explicitly. Full proof in Appendix A.

4. **Experimental result reproducibility.** *Does the paper fully disclose all the information needed to reproduce the main experimental results?* [Yes] All models are public (GPT-2, Pythia, OLMo-2). De-sinking is defined precisely (Algorithm 1). Code will be released.

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